

MATTHIAS DOGBATSEY

Ph.D. Candidate in Mathematics

The University of Alabama
Department of Mathematics
Tuscaloosa, AL
mdogbatsey@crimson.ua.edu
www.mdogbatsey.github.io

EDUCATION

The University of Alabama, Tuscaloosa, AL, United States

- 2027 Ph.D., Mathematics (expected)
Advisors: Prof. Shan Zhao and Prof. Yuanzhen Shao.
- 2024 M.A., Mathematics

European School of Management and Technology (ESMT) Berlin, Berlin, Germany

- 2020 Graduate Certificate in Business Management

African Institute for Mathematical Sciences, Accra, Ghana

- 2020 M.Sc., Mathematical Sciences
Thesis: *Modeling the role of imitation in crime dynamics*.
Advisor: Prof. Farai Nyabadza (University of Johannesburg).

University of Cape Coast, Cape Coast, Ghana

- 2017 B.Sc., Mathematics with Economics

RESEARCH SUMMARY

My research sits at the intersection of scientific computing, numerical analysis, and mathematical biology. In computational biophysics, I develop and analyze numerical methods for the Poisson–Boltzmann theory of molecular solvation, with particular focus on nonrigid protein structures and diffuse-interface formulations of solvation free energy. A parallel strand of my work concerns fast solvers for the nonlinear p -Laplacian operator in three dimensions, contributing both algorithmic innovations and rigorous numerical analysis. More broadly, I apply dynamical systems and compartmental modeling to social phenomena, including crime spread and behavioral imitation, with an eye toward policy-relevant insight. My work is supported by NSF grants DMS-2306991 and DMS-2512104.

RESEARCH INTERESTS

Scientific Computing, Numerical Methods for PDEs, Mathematical Biology, Social Dynamics

RESEARCH EXPERIENCE

The University of Alabama

- 2024 – Present NSF Supported Projects, DMS-2306991, DMS-2512104.
A new diffuse-interface approach to ensemble average solvation energy: modeling, analysis and computation.
 - Numerical solutions to the p -Laplacian operator in 3D.
 - Developed fast solvers for the p -Laplacian operator.
 - Novel diffuse-interface framework for modeling ensemble average solvation energy.

¹Updated May 10, 2026

On calculating polar solvation energies for non-rigid proteins in the Poisson–Boltzmann (PB) theory.

- Analytical treatment of nonrigid structures for the regularized PB models.
- Numerical implementation of PB models incorporating nonrigid solute structures.
- Binding energy implementation for the nonrigid solute structure PB models.

Advisors: Prof. Shan Zhao and Prof. Yuanzhen Shao.

African Institute for Mathematical Sciences

2019 – 2020 Master’s thesis.

Modeling the role of imitation in crime dynamics.

- Developed and analyzed nonlinear compartmental models to study the role of imitation in the spread of crime in Ghanaian society.
- Performed bifurcation analysis and numerical simulations using Python.
- Applied stability theory and parameter sensitivity analysis to assess model behavior under varying assumptions.

Advisor: Prof. Farai Nyabadza (University of Johannesburg).

PUBLICATIONS

Published

2. **M. Dogbatsey**, Z. Chen, Y. Shao, & S. Zhao. *Numerical Solutions to the 3D p -Laplacian Operator*. *Journal of Computational and Applied Mathematics*, 481, 117270 (2026).
1. T. Kwofie, **M. Dogbatsey**, & S. E. Moore. *Curtailling crime dynamics: A mathematical approach*. *Frontiers in Applied Mathematics and Statistics*, 8, 1086745 (2023).

Submitted

2. **M. Dogbatsey**, Y. Shao, E. Alexov, & S. Zhao. *On calculating polar solvation energy of nonrigid proteins in the Poisson–Boltzmann theory*. Submitted (2026).
1. T. Kwofie, **M. Dogbatsey**, A. Adom-Konadu, J. K. K. Asamoah, & S. E. Moore. *Optimal Control and Cost Effectiveness Analysis of Fear in Crime Dynamics*. Submitted (2026).

In Preparation

1. **M. Dogbatsey**, M.B. Uddin, Y. Shao, Z. Chen, & S. Zhao. *Variational Implicit Solvation Model with Size-Modified Poisson–Boltzmann and Diffuse Interface*. In preparation (2026).

TALKS & PRESENTATIONS

6. *Diffuse Interface Variational Implicit Solvation Models with the Size-Modified Poisson–Boltzmann Equation*. AMS 2026 Spring Southeastern Sectional Meeting, Georgia Southern University-Armstrong Campus, Savannah, GA. Mar. 28-29 2026. [Invited]
5. *Numerical solution to the three-dimensional p -Laplace equation*. SIAM Central States Section (CSS 2025), University of Arkansas, Fayetteville, AR. Oct. 11-12 2025. [Invited]
4. *Numerical solution to the three-dimensional p -Laplace equation*. Southeast ACM Graduate Students Workshop, Auburn University, Auburn, AL. Apr. 5-6 2025. [Invited]

3. *Numerical solution to the three-dimensional p -Laplace equation.* SIAM Southeastern Atlantic Section Annual Meeting (SEAS 2025), University of Tennessee, Knoxville, TN. Mar. 21-23 2025. [Contributed]
2. *Assessing the Impact of Intervention Programs on Gang Dynamics: A Mathematical Modeling Approach.* Joint Mathematics Meetings (JMM 2024), Moscone Center, San Francisco, CA. Jan. 3-6 2024. [Invited]
1. *Assessing the Impact of Intervention Programs on Gang Dynamics: A Mathematical Modeling Approach.* Applied Mathematics Seminar, University of Alabama. Dec. 2023. [Invited]

TEACHING EXPERIENCE

The University of Alabama — Instructor of Record

Spring 2026	MATH 237, Introduction to Linear Algebra (two sections).
Summer 2025	MATH 113, Precalculus Trigonometry.
Spring 2025	MATH 227, Calculus III.
Spring 2024	MATH 238, Applied Differential Equations I.
Fall 2023	MATH 115, Precalculus Algebra and Trigonometry (two sections).
Spring 2023	MATH 125, Calculus I.
Fall 2022	MATH 115, Precalculus Algebra and Trigonometry.

The University of Alabama — Teaching Assistant

2021 – 2024	Tutor, MATH 112 through MATH 227.
Spring 2023	Grader, MATH 301, Discrete Mathematics.

SCHOLARSHIPS AND AWARDS

2024 – 2026	Summer Research Fellowship, Department of Mathematics, The University of Alabama.
2024 – 2026	Graduate Student Travel Award, Graduate School, The University of Alabama.
	– AMS Sectional Meeting Travel Support (2026).
	– SIAM CSS 2025 Travel Support (2025).
	– SIAM SEAS 2025 Travel Support (2025).
	– JMM 2024 Travel Support (2023).
2025	SIAM CSS 2025 Travel Award, University of Arkansas.
2025	Southeast ACM Graduate Students Workshop Travel Support, Auburn University.
2025	SIAM SEAS 2025 Travel Award, University of Tennessee, Knoxville.
2022 – 2023	Edith & Richard Ainsworth Endowed Scholarship, Department of Mathematics, The University of Alabama.
2020	Funding to attend Stochastic Processes and Applications Research and Training Workshop, AIMS Ghana.
2019 – 2020	Full scholarship, taught master's degree, AIMS Ghana.
2019	Funding to attend Probability, Analysis and Applications (PAA) Workshop, AIMS Ghana.

SERVICE AND OUTREACH

Departmental and University Service

2024 – 2026	GSA House of Delegates, Delegate. Represented the Department of Mathematics in the Graduate Students Association House of Delegates at the University of Alabama; served on the Academic Development Committee.
-------------	---

Competition Judging and Mentoring

- | | |
|-------------|---|
| 2025, 2026 | SIAM MathWorks Math Modeling (M3) Challenge, Triage Judge.
Evaluated and provided written feedback to teams of high school students on mathematical modeling submissions. |
| 2024, 2025 | SIMIODE Challenge Using Differential Equations Modeling (SCUDEM), Judge.
Assessed undergraduate student research on differential equations modeling in MATLAB. |
| 2022 – 2026 | MATHCOUNTS, Grader.
Graded competition work by sixth- through eighth-grade students in West Alabama. |
| Apr. 2024 | Ohio Science Day, Judge.
Evaluated student research projects for the state of Ohio science competition. |
| Mar. 2024 | UA Undergraduate Research and Creative Activity Conference (URCA), Judge.
Reviewed undergraduate research presentations at the University of Alabama. |

INDUSTRY EXPERIENCE

- | | |
|----------------|---|
| 2026 – Present | Community Engagement Lead , Zappling Learning Inc, Clarksville, TN. |
| 2018 – 2019 | Sales Analyst , Danone SC (FanMilk Plc), Accra, GH. |
| 2017 – 2018 | Assistant Sales Analyst and Sales Operations Officer , Danone SC (FanMilk Plc), Accra, GH. |
| Summer 2015 | Customer Service Intern , Lower Pra Rural Bank Limited, Shama, GH. |

PROFESSIONAL AFFILIATIONS

American Mathematical Society (AMS).
Mathematical Association of America (MAA).
Society of Industrial and Applied Mathematics (SIAM).
National Association of Mathematicians (NAM).
African Society for Biomathematics (ASB).

SKILLS

Programming Languages: Fortran, MATLAB, Python.
Specialist Software: APBS, DelPhi (electrostatics/solvation), VMD (molecular visualization), Git (version control), HPC (slurm).

REFERENCES

Available upon request.